

Edge Ideal and Cover Ideal of a Graph

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Abstract: In this seminar, we explore the connection between combinatorics and algebra through the concepts of edge ideals and cover ideals of a finite simple graph. Let G be a graph with n vertices; then the edge ideal $I(G)$ and the cover ideal $J(G)$ are ideals in the polynomial ring $k[x_1, \dots, x_n]$ where k is a field. The edge ideal $I(G)$ is generated by the monomials corresponding to the edges of G and the cover ideal $J(G)$ is generated by the monomials corresponding to the minimal vertex covers of G . We discuss the primary decomposition of the edge ideals and the cover ideals. In addition, we discuss an algebraic description of the chromatic number of a graph,

$$\chi(G) = \min\{d \mid (x_1 \cdots x_n)^d \in J(G)^d\}.$$